

Supporting Students with a Personal Advisor

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ABSTRACT

Recently, many national and international initiatives show an increasing interest in promoting distance education interventions by Universities. These directives aim at supporting students during their studies providing them with personalized solutions to their problems related not only to the fruition of on-line courses but also to orientation issues. In this paper we present a general architecture of an Embodied Conversational Agent (ECA) that has the main aim to assist students by providing personalized suggestions. This ECA represents the student interface in interacting with a Virtual University environment able to provide different types of services. In particular to support mobility, the ECA architecture has been conceived in order to run also on the student handheld device. The paper discusses the design and technical issues involved in developing this personal agent and the results of an evaluation study aiming at assessing the impact of conversational agents on the effectiveness of the interaction.

Keywords

Personal Advisor, Embodied Conversational Agents, M-learning, Multiagent-Systems, Virtual Universities

Introduction

In this paper we present a general architecture of an Embodied Conversational Agent (ECA) that has the main aim to assist students by providing personalized suggestions. These suggestions can be related not only to the fruition of on-line courses but also to orientation issues regarding student daily life.

The motivations at the basis of this work come from the directive of the Italian Universities that aims at promoting interventions for improving mentoring and orientation services. This directive aims at following students during the entire course of study, giving them personalized suggestions about different topics and helping them to take decisions about their formative process. In particular, taking as a reference the role of an advisor in our Department, his/her main goal consists in assisting and orienting the students focusing on: suggestions about orientation choices, personalization of the study curricula, removal of “obstacles” during the course of study, identification of appropriate and interesting research fields for the thesis, suggestions on how to make students more participative in their formative process. At the first year, each student is assigned to an advisor, chosen between available professors by the Department Council. Each advisor follows a group of students during the entire course of study. After a trial period, we noticed that this service was not fully exploited by students and we investigated the reasons related to this phenomenon. In order to assess the motivations, we made a user study consisting in a questionnaire (150 subjects in total) aiming at understanding which was the user’s expectations about the advisor role and at assessing why students were not using this orientation service.

Analyzing the result of this survey, it came out that the main reason was related to the difficulty to find the advisor always available when needed (professors teach, do research, are involved in meeting, etc.) and also to the shyness of some student that are afraid of asking questions about their problems to professors.

Taking these motivation and the ubiquitous computing vision (Weiser, 1991) into account, we decided to develop VU-MAS, a Virtual University Multi-Agent System (MAS) with the aim of supporting students during

the entire course of study. This system is able to support students by providing a interface to access to several services by combining knowledge stored in distributed sources whenever needed.

Each student can interact with VU-MAS using a personal agent, called MyCoach, represented as an ECA. The main goal of this agent is to monitor the student activities, following his/her learning improvements, but also to select useful material according to the recognized student's goals and needs. The agent is also capable to proactively provide the student with useful suggestions whenever it is needed. As it is designed to run on a smart phone or a PDA, this agent combines e-learning capabilities with mobile computing, thus realizing an m-learning experience where the student can feel always in touch with his advisor.

The paper is structured as follows: after a brief overview on the related work, we will briefly illustrate the VU-MAS architecture, focusing on the description of the interaction aspects of the system. Then we will describe how we extended the Mind-Body architecture, developed in the context of the MagiCster project (de Rosis et al., 2003), in order to implement MyCoach. In order to assess the impact of the conversational agent on the effectiveness of the interaction we performed an evaluation study whose results are reported in the related section. The paper ends with some concluding remarks and some pointers to future work.

Related Work

Developing more human-like systems seems to improve interaction by establishing a more engaging relation with the agent (Dehn and van Mulken, 2000). In this role, ECAs offer to people the possibility to relate with computer media at a social level (Reeves and Nass, 1996) and, therefore, they have potential as facilitators for this type of interventions thus making the interaction more effective and enjoyable (Moreno et al., 2001).

According to research and evaluation studies in the field of intelligent interfaces, ECAs (Gratch et al., 2002) have shown to be a good interaction metaphor when acting in the role of counselors (Marsella, Johnson, & LaBore 2003), personal trainers (Bickmore, 2003), or healthy living advisors (de Rosis et al., 2003). Indeed, ECAs have the potential to involve users in a human-like conversation using verbal and non-verbal signals for providing feedback, showing empathy and emotions in their behavior (Cassell, 2001). Due to these features, an ECA can be successfully employed as interaction metaphor in the pedagogical domain (Johnson, LaBore & Chiu, 2004) and in other domains where is important to settle long term relation with the user (Bickmore & Picard, 2005).

In particular, as far as the interaction with a computational environment supporting the learning process is concerned, the use of conversational agents can be envisaged in the following roles (Roda, Angehrn and Nabeth, 2001): i) as advanced helping facilities ; ii) as personal coaches equipped with specific domain knowledge; iii) as role-playing actors in simulated environments (i.e. 3D virtual worlds). When an ECA acts as a personal coach it has to show a personality appropriate to its role, the ability to guide the user, to reason about user actions, to provide useful suggestions and to communicate through a realistic, interactive interface.

Moreover, even if ECAs have shown to have a good impact on settling an emphatic relation with the user (de Rosis et al., 2005; Cassell et al. 2000), involving them in a deeper and intimate interaction, it is difficult to communicate with these agents whenever needed (i.e. when the user is not in front of a computer but he/she has the need to get suggestions and advices). For example, DESIA (Johnson, LaBore and Chiu, 2004), is a step in this direction. This agent, presented in *Carmen's Bright IDEAS*, has been adapted for running successfully on a handheld device in order to assist in a psychosocial intervention aimed at providing training in problem solving skills (Marsella, Johnson & LaBore, 2003). Therefore, if we want to support student in a continuous way the personal advisor should be available and accessible also on a mobile device. Our work represents a step in this direction.

VU-MAS Architecture Overview

A Virtual University should be able to support students from different points of view, during their formative process. It has to help students not only with online courses and material, but also with logistic information and tuition support. (Cunha, Tavares & Ferreira, 2005). Following this vision, the Virtual University MAS is composed by agents that provide information services of different kind.

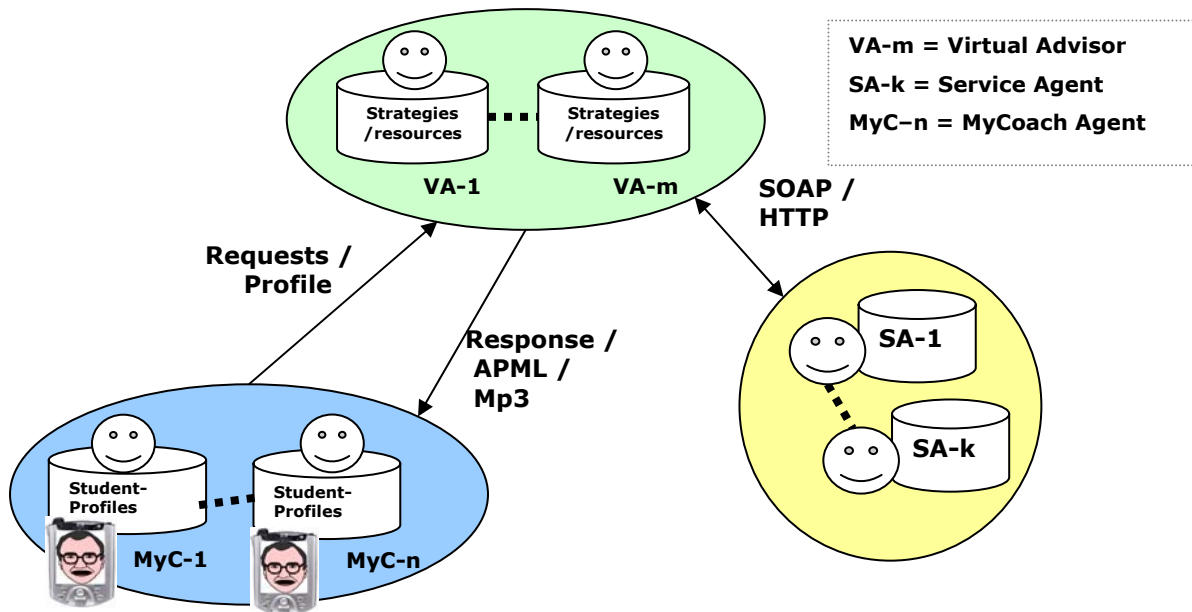


Figure 1. VU-MAS architecture

The agents in VU-MAS are grouped into three main categories: MyCoach, Virtual Advisor and Service Agents.

MyCoaches (MyC-s) are personal agents that manage the student profile and the interaction between the student and his advisor, as we will show in the next section. In VU-Mas there will be a personal agent for each student.

Virtual Advisors (VA-s) have the role of their real world counterpart. In fact, they represent the teachers when they are in the role of advisors. They can access a knowledge base (KB) and use a reasoning engine to decide what to suggest. The KB contains information about the students and the strategy used to advise according to student features. VA-s can store every activity of the students, when they are connected to VU-MAS, either using MyC agents or a web applications, updating the students' profiles accordingly.

Service Agents (SAs) are specialized in providing a specific service, such as administration office, a library or even a teacher, and they can be either simple request/response agents, or more complex ones.

Every human advisor can access and update the KB using his/her VA. He/she is responsible for a set of students. The students use their personal device to connect to VU-MAS. This is a FIPA compliant MAS, in which the communication between MyC and VA agents is made using ACL messages (FIPA,2006). The SAs can be either agents or webservices (Greenwood, Buhler & Reitbauer, 2005). They answer to requests using standard protocols, such as SOAP and HTTP (W3C, 2004).

In VU-MAS, each student is associated to one advisor. The first time the student has to download his MyC agent on his personal device, from the VU-MAS web site. Starting from that moment a Virtual Advisor agent is associated to the student who will be able to interact using MyC ECA and, according to the student year of course, passed exams or specialization level, a stereotype is assigned to him/her. This stereotype is used to set the initial student background knowledge and interests. After the agent initialization, the body of the user stereotype is copied in the student profile, which is updated according to the student interaction with VU-MAS. The profile is used for adapting the information to the student situation and recording the interaction history with the system.

The profile of the students contains personal data (name, birthday, etc.), information about the student curriculum (courses, exams, marks, etc.), preferences, hobbies and interests. Each MyC agent running on the personal device manages and stores the profile of the student in a xml structures, formalized using a common representation language. We decided to use the situational statement language UserML from UbiWorld (Heckmann et al., 2005), since it allows to represent the user model in a very detailed structure, introducing

information useful for the contextualization. Furthermore, as privacy can be a crucial issue in this kind of systems, UserML provides some tags to set privacy retention constraints.

Whenever a MyC agent needs to answer to some questions from the student, if it can be connected to the VU-MAS, it contacts the VA of his advisor. Then MyC sends a request and the student profile to the VA, triggering its tasks. The VA answers sending back the information on the basis of the student profile received, attaching the needed resources. When the student asks some information, the VA looks for information asking to SAs. For example a SA can manage information about courses, such as name, teachers timetables and so on.

MyC agent is supposed to be always connected to VU-MAS. However, if it is not possible to be online, or if the student decides to disconnect, MyC agent has to be able to give some suggestions. In this case, using the natural language generation strategies that will be shown in the next section, it gives an answer reusing information already downloaded on the device during previous sessions, combining also the cached animations and sounds. If these resources are not enough to answer the request, this is stored and the agent will give an answer when connected. Then, when offline, MyC agent can answer to questions that it has already answered to or that have some related links to previous requests, using its cache. In this way the interaction is much less engaging, because the agent cannot exploit new knowledge, but can be used to remember previous answers. This is an issue which will be overcome as well as the computational power increases, or broadband wireless connection become more spread. During the interaction MyCoach keeps tracks of all the questions and answers to update the student profile.

MyCoach: a Personal Advisor Agent

All the interaction described in the previous section is handled by MyCoach agent. This is an Embodied Conversation Agent, with the aim of giving the agent a more engaging look and feel. As mentioned in the introduction, MyCoach agent has been developed as an extension of the architectural schema developed in the context of the MagiCster project. In this project, the ECA was intended as an entity made up of two main components, a '*Mind*' and a '*Body*', interfaced by an I/O language, so as to overcome integration problems and to allow their independence and modularity.

During the interaction, the Agent's Mind has the purpose of deciding what to communicate, by considering the dialogue history, the conversational context and the communicative goal related to the user request. The output of the mind module is an **APML** (Affective Presentation Markup Language) specification of the meaning to be communicated (De Carolis et al., 2003).

The Body has the purpose to interpret and to render this output at the surface level, according to the available communicative channels: different bodies may have different expressive capabilities and therefore may use different channels. This approach was mainly driven by the definition of *communicative function* in Poggi et al. (Poggi et al., 2000). In their theory, a communicative function is a (**meaning**, **signal**) pair, where the **meaning** item corresponds to the communicative value of the **signal** item. For instance, a smile can be the signal of a "joy" emotion.

Therefore, the Mind should convey only the meanings associated with the act to be communicated and the Body should interpret these meanings and render them into an expressive behaviour according to the interaction context and the device used in the interaction.

In the first version of the architecture, the APML interpretation process was embedded in the Body Player (De Carolis et al., 2003). But, since the Agent's believability is strictly related to features such as its personality and role and the cultural and social context, separating the way in which the Body will render that meaning enforces adaptivity.

Mobile devices place limitations on the functionality of an embodied conversational agent since their computational and display capabilities are limited. Let's see now how this architecture structure has been adapted for the development of an ECA that runs also on an handheld computer platform (Figure 2).

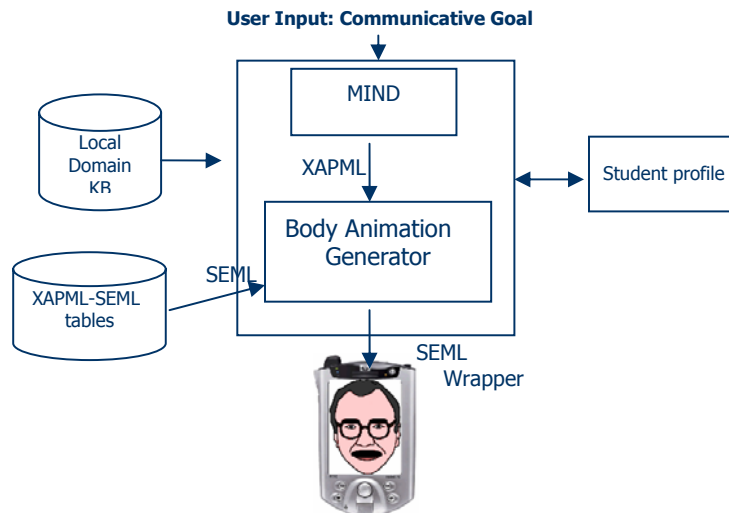


Figure 2. Mind-Body Architecture

Agent's Mind

The task of this module is to generate the specification of the Agent's move at the meaning level. Planning is a task hard to compute, then it is not feasible to have a planner running on an handheld computer, moreover MyCoach has to run on and off-line. For this reason, we adopted a schema based approach.

Given a communicative goal, derived by an interpretation of the user input, the Agent's Mind selects from its plan library the schema or a combination of schemas that allow to satisfy that goal. The result is an XAPML (eXtended APML) specification of meanings to be communicated.

APML is a markup language designed to represent the communicative functions (De Carolis et al., 2003). Each tag of this language is used to add some affective accent to the content of a sentence. For instance the <performative> tag is used to specify what is the communicative speech act of the sentence (Pelachaud and Poggi, 1998).

The extension of APML includes the following information that were not present in the first version: i) besides the specification of what the agent has to communicate it is possible to specify the information **background** (access to more details, visualization of relevant domain objects, etc.); ii) the **focus** attribute of the <performative> tag allows to establish which is the main topic under discussion and to update the student profile with information about what has been communicated to the student; iii) the **voice** attribute has been added since, in case the student interact through its handheld device, the voice output has to be generated on the server side and then passed to the client for being played together with the body animation. These recipes have been designed on the basis of the results of a survey, in which the students were asked to rate a set of information about the department and student life. Figure 3 shows results of second year students.

Analysing the survey results, major interests seem to concern information about modalities for choosing optional exams for completing the third year curricula, accessing student facilities, setting an examination test, get credits and recover from formative debits. Therefore, we developed XAPML schemas for answering to these families of questions and for getting more detailed answers once selected a topic question. These schema allows to organize the sequencing of the answer and to manage the multimodality of the interaction, combining text, voices, and expressions of the face.

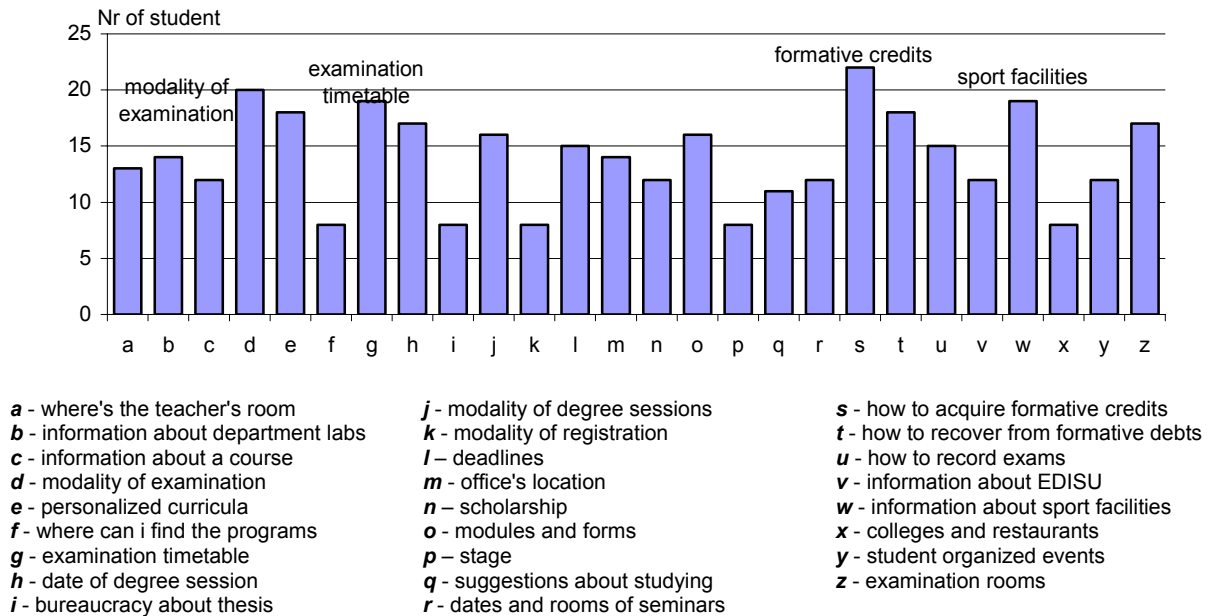


Figure 3. Results of the questionnaire of second year students

The following is a recipe for providing personalized suggestions about “optional courses”:

```
<XAPML-recipe goal="suggestion-choose-course">
  <agent>
    [FORALL optional_course(topic-i) DO
      IF NOT(HasInterest(S,topic-i)) DO
        [
          <performative type="inform" affect="sorry-for" voice="sorry.mp3"> I'm sorry for
            you but</performative>
          <performative type="suggest" focus="optional_course(topic-i)" voice="course-
            i.mp3"> I have to suggest you to follow the course [topic-i]</performative>
          <performative type="inform" voice="motivation-i.mp3"> even if I know you don't like
            [SuperClass(topic-i) OR Related(topic-i)]</performative>
        ]
      IF (HasInterest(S,topic-i))
        <performative type="inform" affect="happy-for" voice="happy.mp3"> I'm happy
          </performative>
        <performative type="suggest" focus="optional_course(topic-i)" voice="course-
          i.mp3"> to suggest you to add the course of [topic-i] to your curriculum
          </performative>
        <performative type="inform" voice="motivation-i.mp3"> since I know that you like
          [SuperClass(topic-i)OR Related(topic-i)]</performative>
      ]
    ]
  </agent>
  <background>
    [FORALL optional_course(topic-i) DO
      IF detail <object type="link" label="[topic-i]">[URI-topic-i]</object>
    ]
  </background>
</XAPML-recipe>
```

These schemas are applied by the Mind in order to generate the instantiated XAPML specification. In the previous example, square parenthesis represent instructions to the XAPML generator that will substitute them with the appropriate text by manipulating information in the Domain Knowledge Base. This transformation is executed on the server side and not on the handheld computer where, as we will see later on, only instantiated XAPML specification are interpreted and rendered at a surface level.

Agent's Body

When the student downloads the agent from the Department Web Site she can choose its Body according to his/her preferences. The selected body, obviously, influences the way in which the agent is able to express meanings.

Since our main aim was not to develop a new ECA player but to test the validity of our approach, at the moment we have developed only two types of bodies (Table 2): a face (male professor and a young female professor assistant) and a character (a funny ball). This last one could be used, for instance, for interacting with children.

Each ECA conveys the same meanings using signals that are typical of its body. For instance the two faces will use as signal channels those typical of a face (eye, eyebrow, mouth, gaze, etc.) while the funny ball will use movements, colors and changes in its dimensions as communicative channels. The XAPML specification is not directly interpreted by the player that is the same for all the different bodies. This is possible since the proposed architecture decouples meanings from signals.

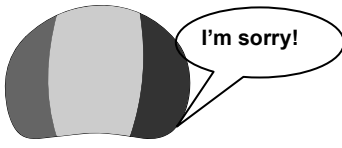
Each body has a conditional meaning-signal table that allows to appropriately translate an XAPML tag into a Signal Expression Markup Language (SEML). SEML tags define the expressions that can be performed on each channel of the Body. Table 1 specifies how a body's `<act>` can be specified in SEML along the appropriate channels for the *face* (Gratch et al., 2002; Poggi et al., 2000) and the *ball*.

Table 1. SEML signals expression for two different bodies

Face Body	Funny Ball
<code><play channel="eye" value="" /></code> <code><play channel="eyebrow" value="" /></code> <code><play channel="mouth" value="" /></code> <code><play channel="gaze" value="" /></code> <code><play channel="voice" value="" /></code> <code><play channel="text" value="" /></code>	<code><play channel="shape" value="" /></code> <code><play channel="color" value="" /></code> <code><play channel="size" value="" /></code> <code><play channel="look" value="" /></code> <code><play channel="sound" value="" /></code> <code><play channel="text" value="" /></code>

An example of SEML output for both the “professor” and the “funny ball” bodies, deriving from a transformation of a XAPML performative, is shown in Table 2.

Table 2. Example of transformation from XAPML to SEML

XAPML:	
<code><performative type="inform" affect="sorry-for" voice="sorry.mp3">I'm sorry for </performative></code>	
The professor:	The funny ball:
<code><act></code> <code><play channel="mouth" value="serious"/></code> <code><play channel="voice" value="sorry.mp3"/></code> <code><play channel="text" value="I'm sorry!"/></code> <code></act></code>	<code><act></code> <code><play channel="shape" value="deflated"/></code> <code><play channel="color" value="grey"/></code> <code><play channel="size" value="1"/></code> <code><play channel="sound" value="cry.mp3"/></code> <code><play channel="text" value="I'm sorry!"/></code> <code></act></code>
	

The first row shows the XAPML input, the second one the SEML specification for the two bodies and the third one their rendering by the player. In this example, the face body will show a typical human-like expression of *sorry-for*, while the funny-ball will express it by changing its shape (deflated) and color (grey instead of its normal bright colors).

An Example of Interaction

VU-MAS allows to access different types of information and services and can be used as an interface for online courses. Students can interact with it using MyCoach as an interface. According to the type of student request the agent will adapt the presentation to the student's profile, which contains his/her preferences, skill level, progress in attending a course, etc.

In order to show an example of interaction, let us assume, as a scenario example, the following one:

Vincenzo, a second year student, needs some help in choosing some optional exams to organize his study curriculum. His personal profile, besides his personal data, contains the information about his student curriculum (given exams and relative marks), interests and preferences. In particular, analysing these data, it is possible to infer that he is interested in Videogames and that he is skilled in Programming while he is not very good in Mathematics.

Therefore, MyCoach has to suggest him the courses that, according to Vincenzo's profile, are the most appropriate to his skills and previous years curriculum.

Once the personal advisor is activated by the student, it greets him and prompts a list of tasks on which it could help (Figure 4a). From available options the student chooses the course planning help, by clicking on the relative link. The possible choices are planning of the *official* and *personalized* curricula. Let's suppose that the student selects the personalized curriculum option (Figure 4b). Obviously, the student can insert a free text question using the keyboard icon if the list of options does not contain what he was looking for.



Figure 4. MyCoach a) greeting the student, b) proposing help about completing the third year curriculum

MyCoach will then contact the VA responsible for Vincenzo and will send to this agent the request and the statements in the student profile that are relevant to this goal. In this scenario the triggered communicative goal will be "Suggest(MyCoach,Vincenzo,Choose-Course)" and therefore the corresponding XAPML schema shown in the previous Section. This schema will be instantiated then with a list of courses that fit the user profile. To this aim, the VA asks to the SA responsible for curriculum planning the list of possible courses that will be extracted from the related KB.

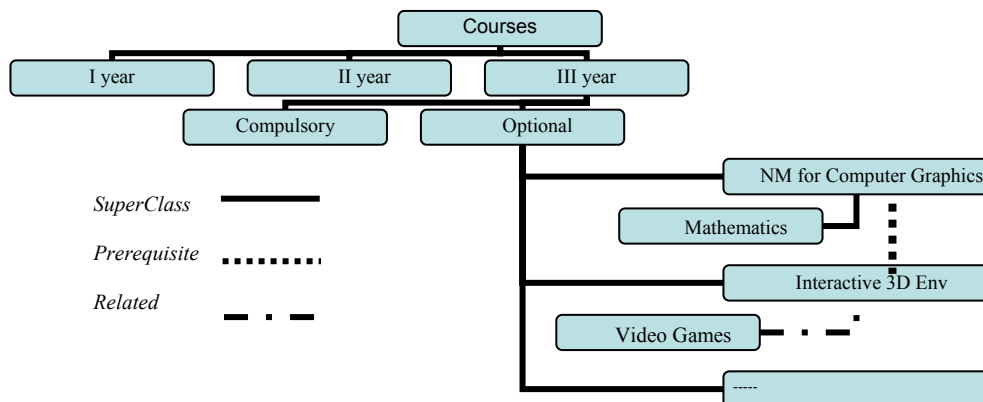


Figure 5. Diagram describing the structure of the Courses KB.

Figure 5 illustrates a simple diagram representing the Courses KB. As we can see from the diagram, and considering the information in the student profile, a list of possible courses is determined. Thereafter, MyCoach suggests a list of possible courses to insert in the curriculum as shown in Figure 6.

In the given scenario, the course *Numerical Methods for Computer Graphics* is considered useful, because it is a prerequisite for the course of *Interactive 3D Environment*, which is connected to the student interests in *Videogames*. However, since the student has got a bad mark in *Mathematics*, therefore this is not one of his preferred topics, the expression of the body of the agent for the first proposal is a sad face (Figure 6 a) and b)), while the second suggestion is communicated in a happy mode (Figure 6 c) and d)). The instantiated XAPML output generated for the illustrated example is the following:

```

<XAPML>
  <agent>
    <performative type="inform" affect="sorry-for" voice="sorry.mp3"> I'm sorry for
    you but</performative>
    <performative type="suggest" focus="optional_course(Numeric Methods for Computer
    Graphics)" voice="NMCg.mp3"> I have to suggest you to follow the course of Numeric
    Methods for Computer Graphics </performative>
    <performative type="inform" voice="motivation-NMCG.mp3"> even if I know that you
    don't like Mathematics</performative>
    <performative type="inform" affect="happy-for" voice="sorry.mp3"> I'm
happy</performative>
    <performative type="suggest" focus="optional_course(Interactive 3D Environment)"
    voice="I3DE.mp3"> to suggest you to add Interactive 3D Environment to your
    curriculum</performative>
    <performative type="inform" voice="motivation-I3DE.mp3"> since I know that you
    really like VideoGames</performative>
  </agent>
  <background>
    <object type="link" label="NM for Computer Graphics">LINK</object>
    <object type="link" label="Interactive 3D environments">LINK</object>
    <object type="link" label="Computer Graphics">LINK</object>
    <object type="link" label="More...">LINK</object>
  </background>
</XAPML>

```

Figure 6 shows how the schema is played by the agent's body when suggesting these courses to the student.

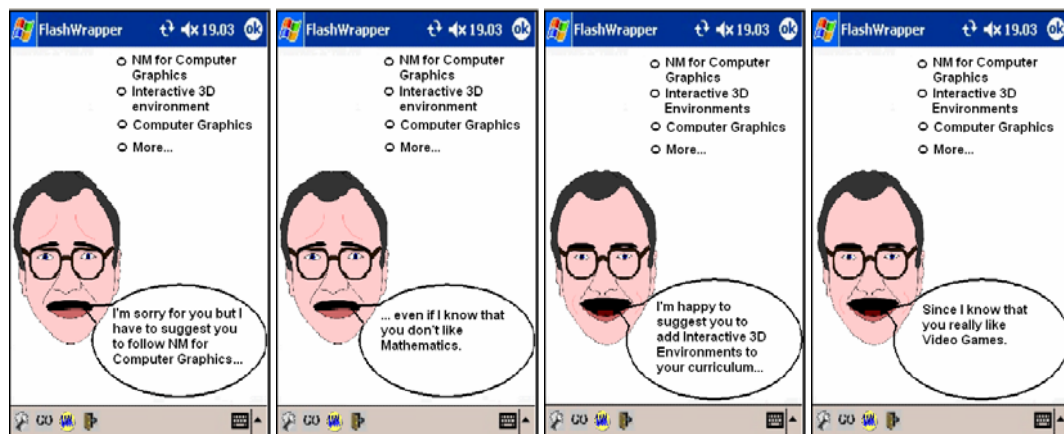


Figure 6. a) and b) "happy-for" expression; c) and d) "sorry-for" expression

The student can always access his profile to modify his preferences or updating his curriculum clicking on an icon in the lower bar. When the user searches for details in a different page, it is always possible to go back and talk to the agent by selecting the icon on the bottom status bar of the handheld application. Moreover it is possible to change the attitude of the advisor, selecting different faces.

Evaluation

Recently embodied conversational agents have been widely used in e-learning projects, even if there are no proved evidence that the use of an ECA can improve the learning experience. To support our proposed approach we have developed a working prototype and we have conducted some evaluation studies with the main goal of assess the effectiveness of the system, in term of level of believability and reliability, perceived by the students.

Moreover, this study was aiming also at assessing the student attitudes toward the agent in terms of how much the presence of an embodied agent improved the effectiveness and engagement of interaction. This has been evaluated through the use of a questionnaire aiming at understanding how much the student liked the character and how much they found helpful MyCoach suggestions.

To this aim we have conducted a test with two groups of ten students of the second year of our Department. Both groups had to execute the task of planning their courses for the third year.

Figure 7. User Profile interface

The first group was asked to do it using a PDA and MyCoach personal advisor as interface. The second group was asked to perform the same task using the web pages of our Department. These are mainly a collection of printable files, so they can be considered similar to a paper guide.

In order to give to MyCoach the capability to provide personalized suggestions we asked to the students belonging to the first group first to create their profiles using a simple user interface (Figure 7) and then to interact with MyCoach.

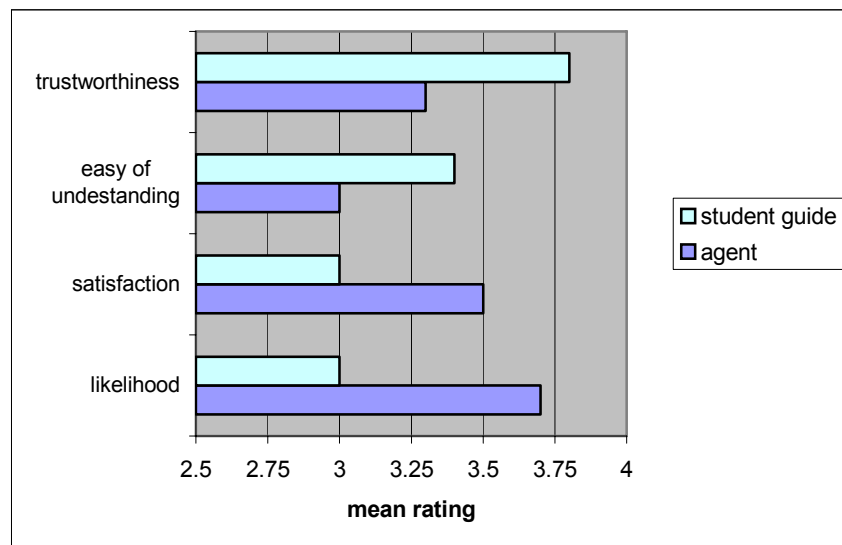


Figure 8. Comparison of features between agent and students webpages

After the experiment, a *post-test questionnaire* (Appendix 1) was administered to measure the impact of the suggestions about courses on participants' subsequent choice. This comprised a range of different measures concerning both informational and interaction aspects. To assess the effectiveness of the 'information provision' aspect of the message, we included ratings of satisfaction and ease of understanding (both on five-point unipolar Likert scales). In addition for the first group, questions about reliability and believability of the ECA were added.

For comparing the data obtained from questionnaires we used t-test. As it could be seen from Figure 8, the level of trustworthiness is lower for MyCoach when compared to webpages of the department. The difference between means is statistically significant for $p=0,05$ where t-test value is $t_1(18)=-1,98$. Moreover the 'easy of understanding' feature is higher for written text in the website. It is confirmed by t-test with $t_2(18)=-1,5$. However difference between means becomes statistically significant for $p=0,1$. In our opinion this can be due to the fact that subjects had more time for reading written information whereas MyCoach advices were spoken. However, the grade of satisfaction and likelihood is higher for the agent. The t-test values are $t_3(18)=1,63$ and $t_4(18)=2,69$ respectively. The first feature is statistically significant at $p=0,1$ level and the second at $p=0,05$ level. In fact, the webpages of our department do not permit personalization, but contains a static list of course names without any link to student preferences or attitudes.

As mentioned earlier, the questionnaire given to the first group of students contained a set of items to measure participants' perceptions towards the agent. Participants were requested to rate the agent along the following dimensions: helpfulness, likeability, reliability and competency. All ratings were made using five-point unipolar likert scales.

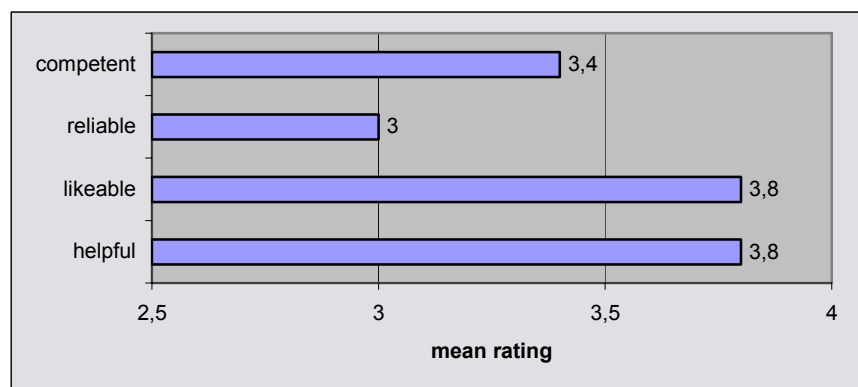


Figure 9. Measure of some dimension of the agent behaviour

The graph (Figure 9) shows that the reliability level has been evaluated with a lower score, even if the students considered the use of the personal agent helpful and likeable.

Students think that the agent is helpful even if they showed to trust more paper based information. Unfortunately the reliability and trustworthiness of a system like this often relays on a long term experience. In our tests users interacted with the agent for a short period of time, which perhaps was not enough to create this relationship. In this view, improving the social and affective behaviour of the agent might lead to a higher level of reliability and trust.

Implementation Issues

VU-MAS is a MAS developed using JADE, a FIPA compliant framework to build multi agent system in java (JADE 2006), and its extension LEAP for handheld devices. MyCoach is one of the agents in this MAS. The Service Agents can be either agents or web-services, which can be connected to the MAS using a gateway (Greenwood, Buhler & Reitbauer, 2005).

MyCoach is implemented combining Macromedia Flash ActionScript with the .NET Compact Framework using Visual Basic.

The interaction may be performed on or off-line; when the user is on-line, the computation of the MIND output is performed server side and the rendering on the handheld (client-side). In this case, when the user connects to the server and asks for specific information the related communicative goal is passed to the system that computes and sends back the XAPML specification of the communication, the mp3 file corresponding to the voice output generated by a TTS on the server side and the set of frames needed for that animation if not yet available on the client side. When the agent is off-line uses pre-stored data to answer to questions, as described in the section about the architecture.

On the client side, MyCoach composes and plays a combination of animation and sound resources loaded onto the Pocket PC.

As in DESIA, with whom this system has a lot of similarities from the implementation point of view, we store the different animations for each body channel. A scene is a collection of sequences of animation frames that are composed in real time as needed according to the SEML specification. Obviously, one SEML specification is then translated into a sequence of frames that are synchronized with the mp3 files by the SEML-Wrapper Module.

Conclusions

MyCoach is a personal agent integrated into a Virtual University MAS. This is a system that has to support the students along their studies by providing orientation suggestions and information useful to solve their problems. The agents cooperate together to reach the main goal of finding useful information, which seems to be one of the most important issues to improve the student study career.

MyCoach is an animated agent designed to be consulted on a handheld device. For this reason we decided to use an extension of the Mind-Body architecture already used in the context of the MagiCster project. In particular, the Mind, running on the server side, decides what to say and generates, through the use of plan “recipes”, a specification (in XAPML) of the meaning to be communicated by the Body that can be used to generate the agent behaviour on the handheld computer (client side). Since the architecture can support the use of different “bodies”, that can express meanings using different signals according to their available channels, a meaning-signal table specifies how a specific body has to convey particular meanings. This is represented through the use of another markup language SEML that specify which combination of signal to employ in correspondence of each XAPML tag. Then, the SEML specification is interpreted by a Wrapper, developed in Flash for MyCoach application, that is responsible for the synchronization and the rendering of the agent animations.

We are aware that MyCoach agent animations are very simple, but this was a first step in order to test how the architecture could support this kind of interaction.

The evaluation studies results are promising, so we are planning to develop more characters with different bodies and interaction attitudes in order to let a higher level of personalization. This is important not only for practical reason, but also because the more the agent can be personalized the more engaging the interaction could be. In particular, we plan to add to MyCoach mind the capability to use motivational and affective strategies to encourage the student when needed.

Further evaluation studies will be done to investigate other features of the system. For example it could be interesting to compare the suggestions of our personal agent with those of a real advisor.

References

- Bickmore, T. (2003). Relational Agents: Effecting Change through Human-Computer Relationships. *PhD Thesis*, Media Arts & Sciences, Massachusetts Institute of Technology.
- Bickmore, T. W., & Picard, R. W. (2005). Establishing and maintaining long-term human-computer relationships. *ACM Transactions on Computer Human Interaction*, 12 (2), 293-327.
- Cassell, J., Sullivan, J., Prevost, S., & Churchill, E. (2000). *Embodied Conversational Agents*, Cambridge, MA: MIT Press.

- Cassell, J. (2001). Embodied conversational agents: representation and intelligence in user interfaces. *AI Magazine*, 22 (4), 67-83.
- Cunha, M. M., Tavares, A. J., & Ferreira, L. (2005). Infrastructures for the Virtual University, EUNIS 2005 - Leadership and Strategy in a Cyber-Infrastructure World, 21-24 June 2005, Manchester, United Kingdom.
- De Carolis, B., Pelachaud, C., Poggi, I., & Steedman, M. (2003). APML, a Markup Language for Believable Behavior Generation. In H Prendinger and M Ishizuka (Eds), *Life-like Characters. Tools, Affective Functions and Applications*, Berlin: Springer.
- de Rosis, F., Cavalluzzi, A., Mazzotta, I., and Novielli, N. (2005). Can embodied conversational agents induce empathy in users? *AISB'05 Virtual Social Characters Symposium*, 12-15 April 2005, Hatfield, United Kingdom.
- de Rosis, F., De Carolis, B., Carofiglio, V., & Pizzutilo, S. (2003). Shallow and inner forms of emotional intelligence in advisory dialog simulation. In H Prendinger & M Ishizuka (Eds.), *Life-like Characters. Tools, Affective Functions and Applications*, Berlin: Springer.
- Dehn, D. M., & van Mulken, S. (2000). The impact of animated interface agents: a review of empirical research. *International Journal of Human Computer Studies*, 52 (1), 1-22.
- Ekman, P. (1993). Facial expression of emotion. *American Psychologist*, 48, 384-392.
- FIPA Home Page (2006), <http://www.fipa.org>.
- Gratch, J., Rickel, J., Andre, J., Badler, N., Cassell, J., & Petajan, E. (2002) Creating Interactive Virtual Humans: Some Assembly Required. *IEEE Intelligent Systems*, July/August, 54-63.
- Greenwood, D., Buhler, P., & Reitbauer, A. (2005). Web Service Discovery and Composition using the Web Service Integration Gateway. Proceedings of the 2005 IEEE International Conference on e-Technology, e-Commerce and e-Service, 29 March – 1 April 2005, Hong Kong.
- Heckmann, D. (2003). Introducing Situational Statements as an integrating Data Structure for User Modeling, Context-Awareness and Resource Adaptive Computing, *ABIS 2003*, Karlsruhe, Germany, 283-286.
- Heckmann, D., Schwartz, T., Brandherm, B., Kröner, A. (2005). Decentralized User Modeling with UserML and GUMO. *Proceedings of the Workshop on Decentralized, Agent Based and Social Approaches to User Modelling (DASUM 2005)*, Edinburgh, Scotland, 61-65.
- JADE (2006). *Java Agent Development Framework*, <http://jade.cselt.it/>.
- Johnson, W. L., LaBore, L., & Chiu, Y. C. (2004). A Pedagogical Agent for Psychosocial Intervention on a Handheld Computer. *AAAI Fall Symposium on Dialogue Systems for Health Communication*, October 22-24, 2004, Washington DC, USA.
- Marsella, S. C., Johnson, W. L., & LaBore, C. M. (2003). Interactive pedagogical drama for health interventions. In U. Hoppe et al. (Eds.), *Artificial Intelligence in Education: Shaping the Future of Learning through Intelligent Technologies*, Amsterdam: IOS Press, 341-348.
- Moreno, R., Mayer, R. E., Spires, H., & Lester, J. (2001). The case for social agency in computer-based teaching: Do students learn more deeply when they interact with animated pedagogical agents? *Cognition and Instruction*, 19, 177-213.
- Pelachaud, C., Carofiglio, V., De Carolis, B., de Rosis, F., & Poggi, I. (2002). Embodied Contextual Agents for Information Delivery Applications. *Proceedings of AAMAS'02*, July 15-19, 2002, Bologna, Italy.
- Pelachaud, C. (2005). Multimodal expressive embodied conversational agents. In *Proceedings of the 13th Annual ACM international Conference on Multimedia*, ACM Press, New York, NY, 683-689.
- Pelachaud, C., & Poggi, I. (1998). Multimodal communication between synthetic agents. In *Proceedings of the Working Conference on Advanced Visual interfaces*, May 24-27, 1998, L'Aquila, Italy.

Poggi, I., Pelachaud, C., & de Rosis, F. (2000). Eye communication in a conversational 3D synthetic agent. *AI Communications*, 13 (3), 169-181.

Reeves, B., & Nass, C. (1996). *The Media Equation*, New York: Cambridge University Press.

Roda, C., Angehrn, A., & Nabeth, T. (2001). Conversational Agents for Advanced Learning: Applications and Research. *Proceedings BotShow 2001*, June 14, 2001, Paris, France.

W3C Web Services Architecture Working Group Note (2004). <http://www.w3.org/TR/ws-arch/>.

Weiser, M. (1991). The Computer for the 21st Century. *Scientific American*, September, 94-104.

Appendix 1

Personal Information:

- a) Are you a student of: first year[] second year [] third year []
b) Age: _____ c) Sex: _____

Please answer this questions:

1) On the basis of what you have just heard how likely are you to follow the recommendations for planning your courses?

1	2	3	4	5
Not at all				Extremely
likely				likely

2) How satisfied were you with the suggestion contained in the message?

1	2	3	4	5
Not at all				Extremely
satisfied				satisfied

3) How easy did you think that the message was to understand?

1	2	3	4	5
Not at all				Extremely
easy				easy

4) How trustworthy did you think the content of the message was?

1	2	3	4	5
Not at all				Extremely
trustworthy				trustworthy

(Only for the group using the Agent Modality)

For each word below, please indicate how well it describes the computer character that you have just seen. Note that you are now evaluating the character, NOT the message.

a) Helpful

1	2	3	4	5
Not at all				Extremely

b) Likable

1	2	3	4	5
Not at all				Extremely

c) Reliable

1	2	3	4	5
Not at all				Extremely

d) Competent

1	2	3	4	5
Not at all				Extremely

End of questionnaire.